

ME 1411: Material Science

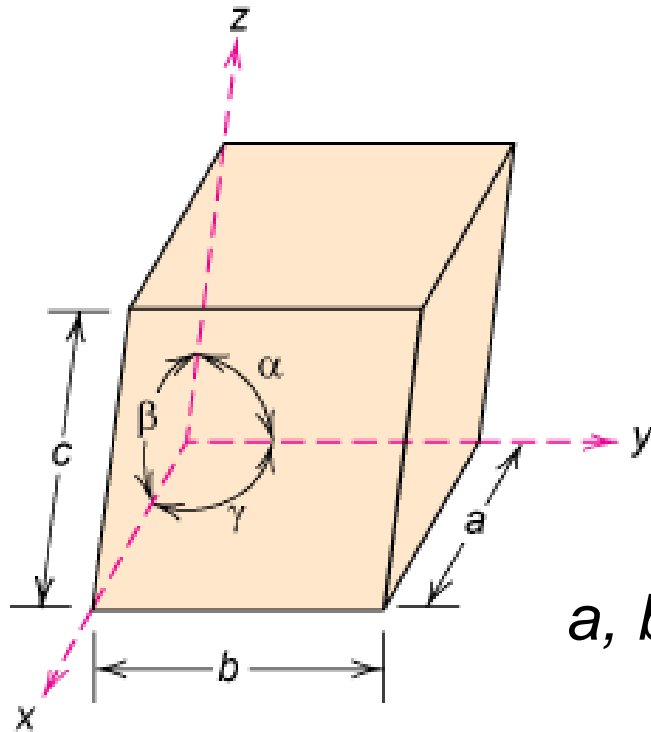
Crystallographic Directions and Planes

Coordinates - René Descartes and the Fly on the Ceiling



Crystal Systems

Unit cell: smallest repetitive volume which contains the complete lattice pattern of a crystal.

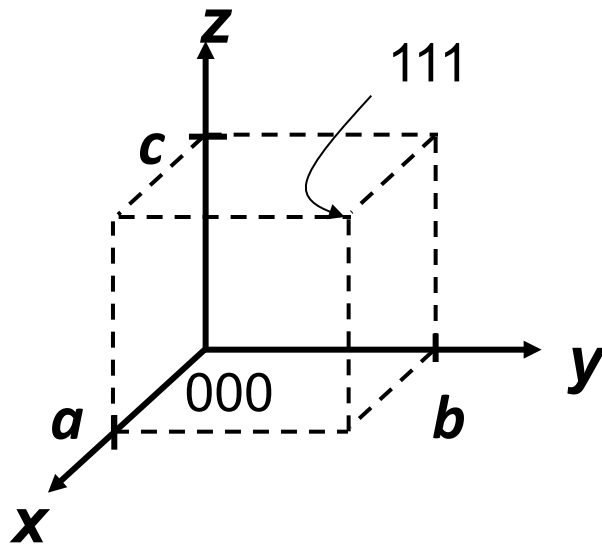


a , b , and c are the lattice constants

Point Coordinates

Point coordinates for unit cell center are:

$$a/2, b/2, c/2 \quad \frac{1}{2} \frac{1}{2} \frac{1}{2}$$



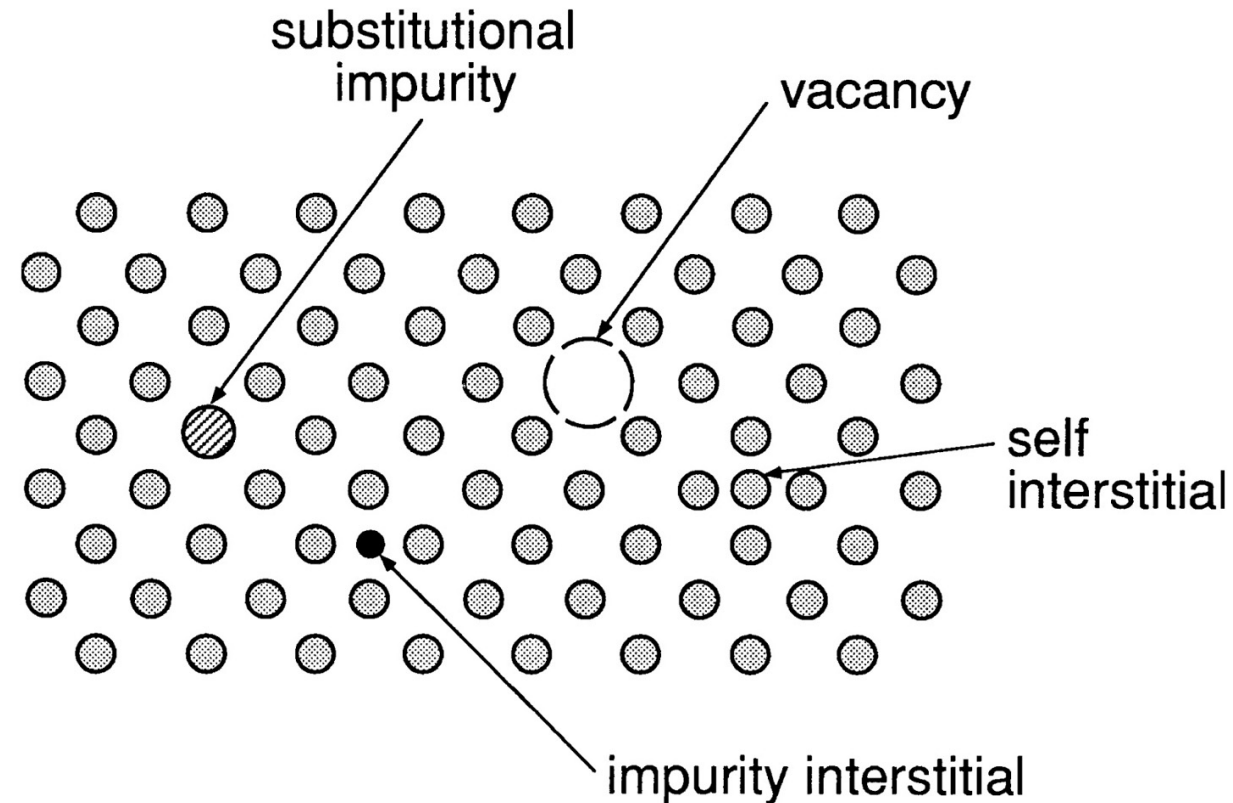
Point coordinates for unit cell corner are

$$111$$

Notice: point coordinates are not in brackets, no commas

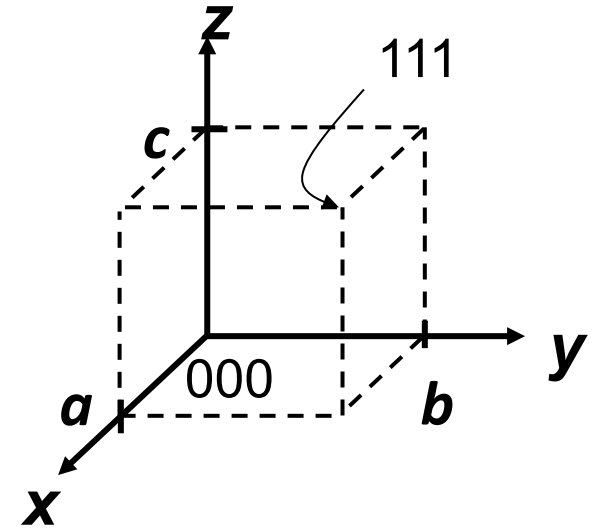
Point Coordinates help us locate defects

Point Defects



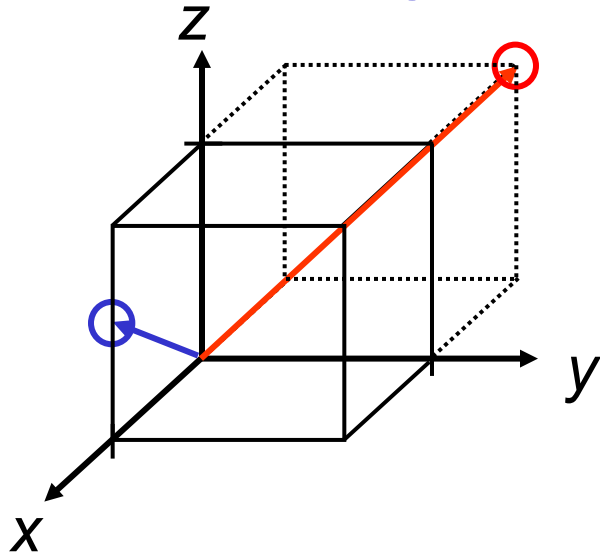
Examples

1. Point coordinates for Point A are?
2. Point coordinates for Point B are?



Crystallographic Directions-Miller Indices

Algorithm



1. Find point coordinates for the tail and tip (Read off projections in terms of unit cell dimensions a , b , and c)
2. Tip – tail
3. Adjust to smallest integer values
4. Enclose in square brackets, **no commas**
 $[uvw]$
5. For **negative value**, put a bar over the index.

ex: $1, 0, \frac{1}{2} \Rightarrow 2, 0, 1 \Rightarrow [201]$

$-1, 1, 1 \Rightarrow [\bar{1}11]$ where overbar represents a negative index

Notice: Directions have square brackets

Miller's Indices allow us to describe dislocations

Edge Dislocation

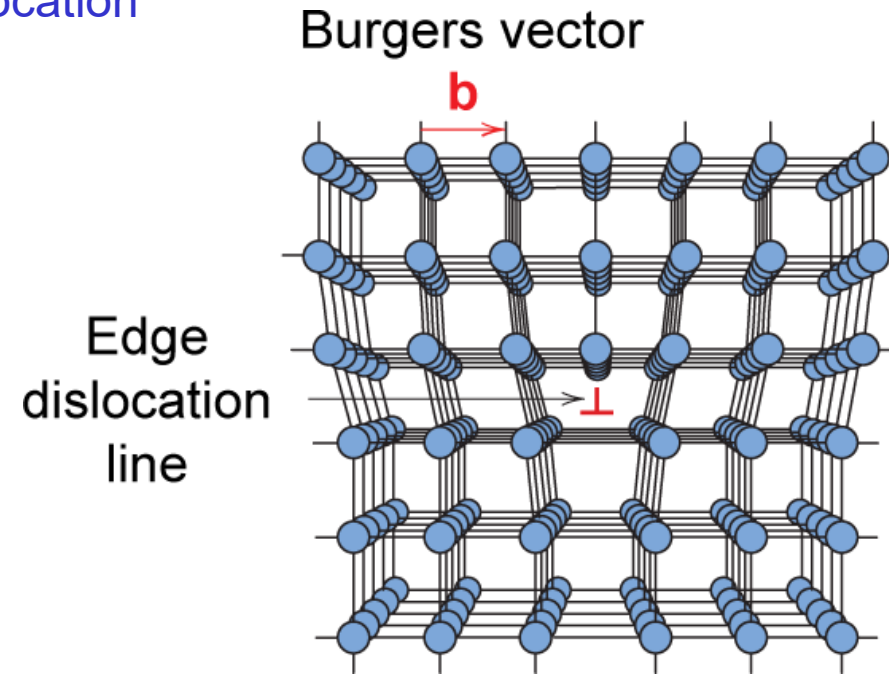
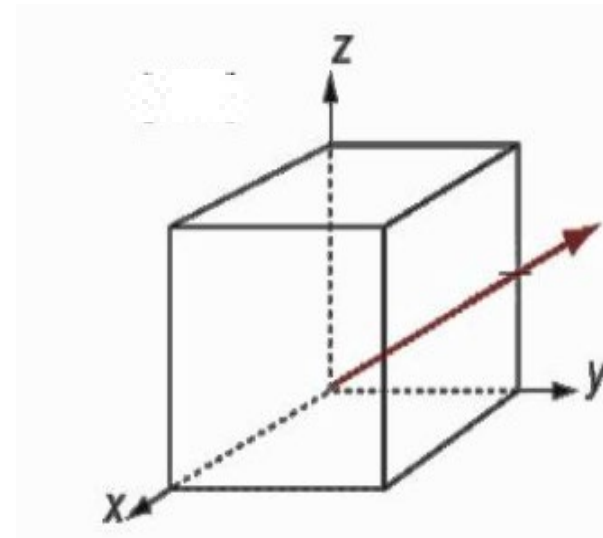


Fig. 5.8, Callister & Rethwisch 3e.

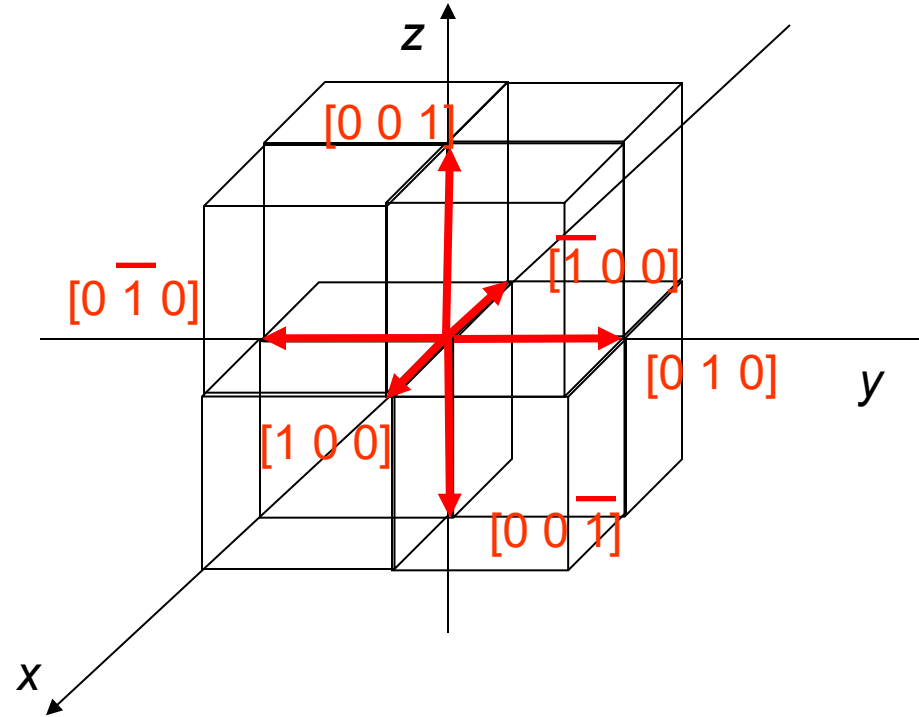
Examples

1. The direction indices of a direction from one point $(3/4, 0, 1/4)$ to another point $(1/4, 1/2, 1/2)$ in a BCC unit cell are?
2. The direction indices of a direction in a BCC unit cell as shown are



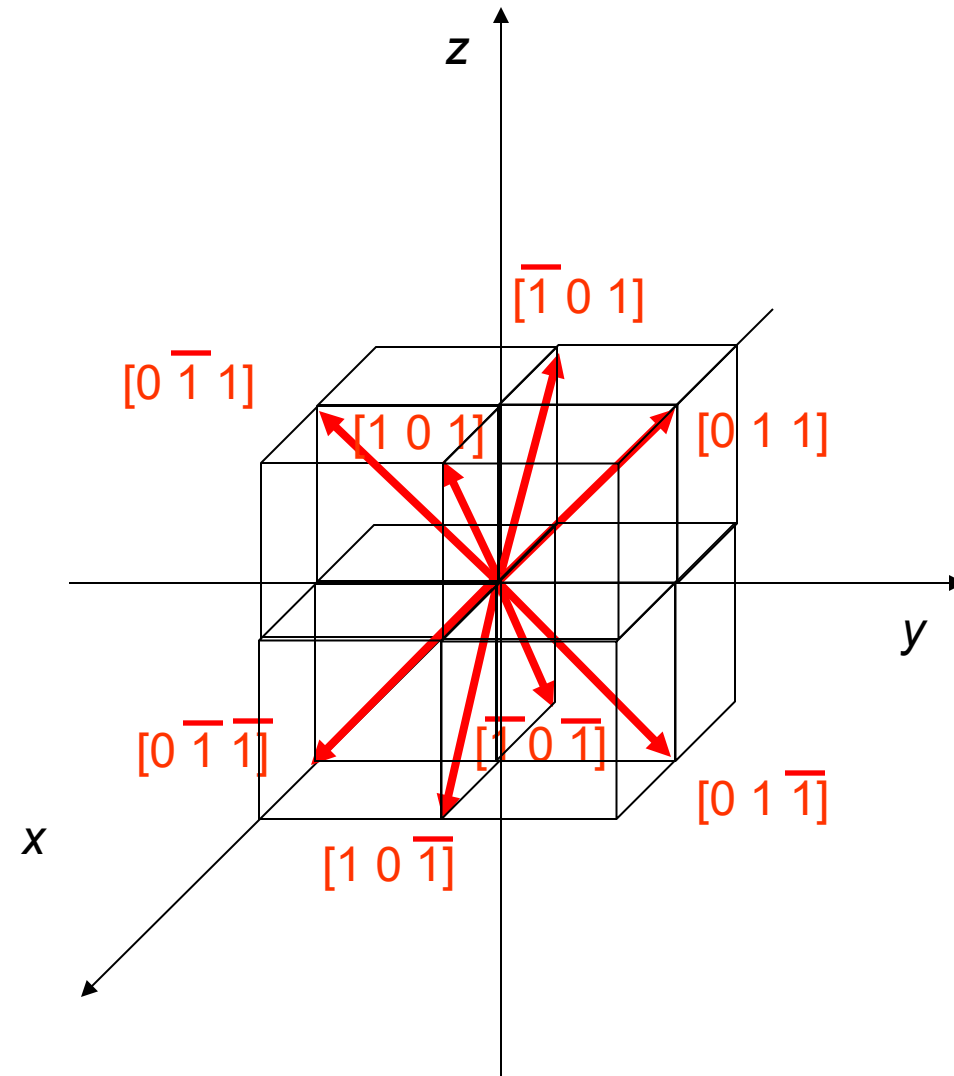
Families of directions $\langle uvw \rangle$

Several nonparallel directions with different indices are crystallographically equivalent.



Cubic Unit Cell $\langle 1\ 0\ 0 \rangle$

Notice: Families of directions have angle brackets

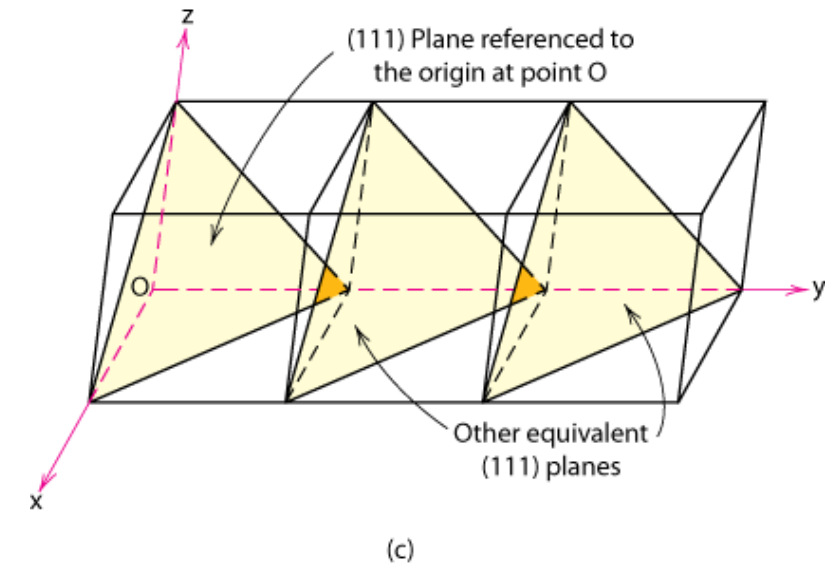
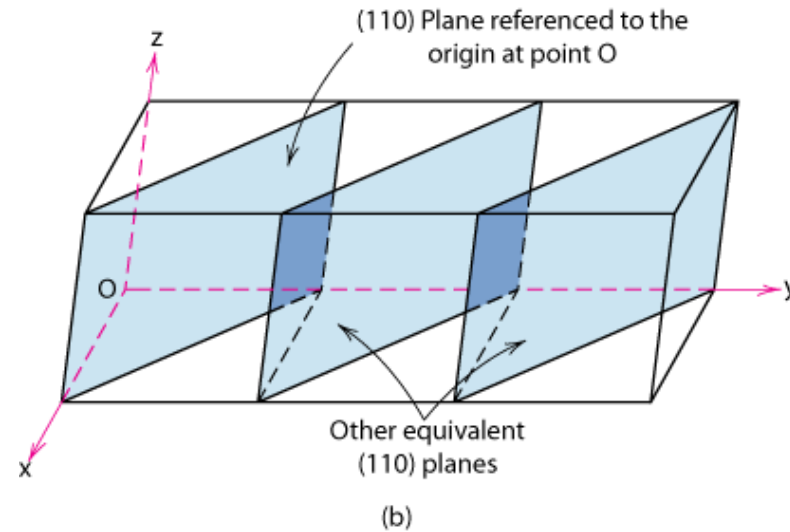
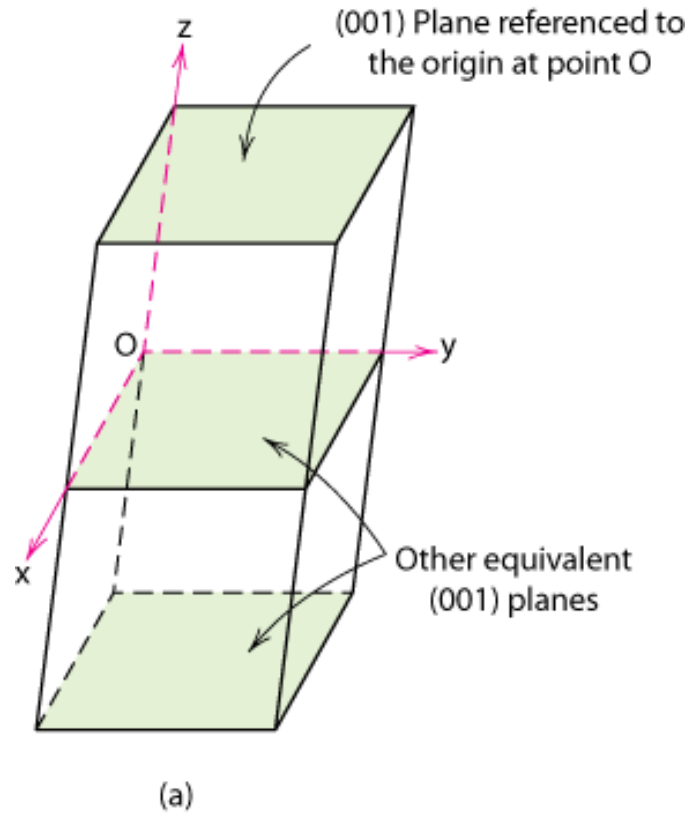


Cubic Unit Cell $\langle 0\ 1\ 1 \rangle$

Crystallographic Planes--Miller Indices

- Miller Indices:
 - Reciprocals of the (three) axial intercepts for a plane, cleared of fractions & common multiples.
 - All parallel planes have same Miller indices.
- When calculating the Miller Indices, choose a plane **NOT** passing through the origin.

Crystallographic Planes



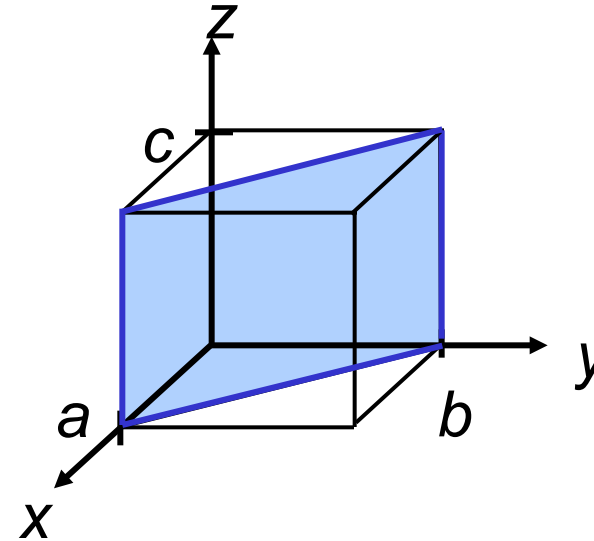
Process for determining Miller Indices of Planes

1. If plane passes through the origin, choose a parallel plane
2. Find the(three) axial intercepts (a,b,c) for a plane.
3. Take the reciprocal of each
4. Reduce to the lowest integers cleared of fractions
5. Millers indices use parentheses

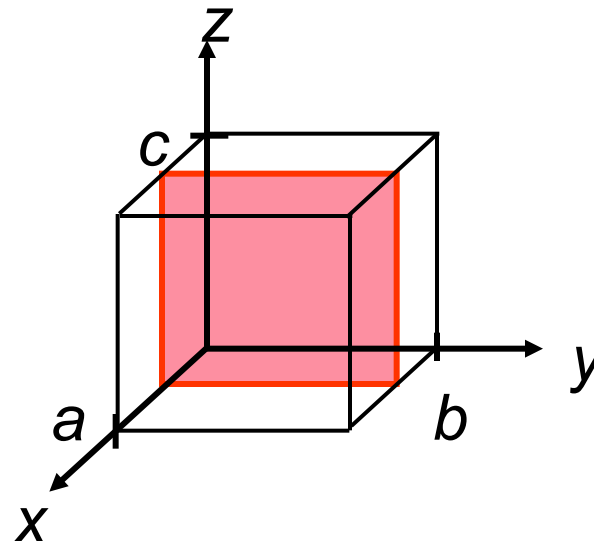
Ex: (111), (110), etc.

Crystallographic Planes

<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	1	1	∞
2. Reciprocals	1/1	1/1	1/ ∞
	1	1	0
3. Reduction	1	1	0
4. Miller Indices	(110)		



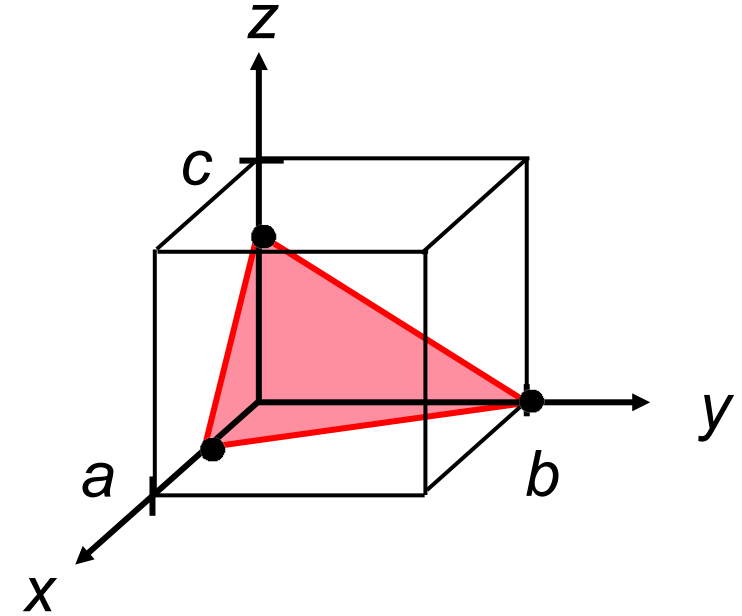
<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	1/2	∞	∞
2. Reciprocals	1/(1/2)	1/ ∞	1/ ∞
	2	0	0
3. Reduction	1	0	0
4. Miller Indices	(100)		



Notice: Planes have parentheses

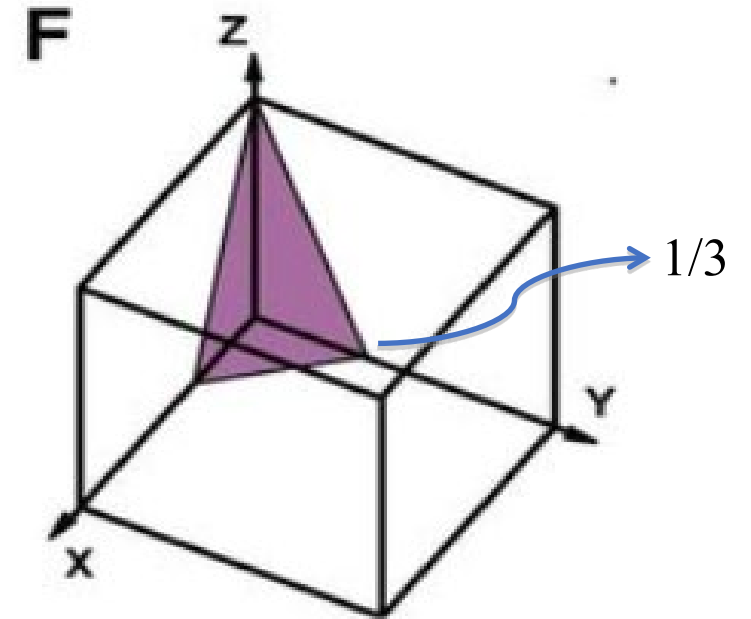
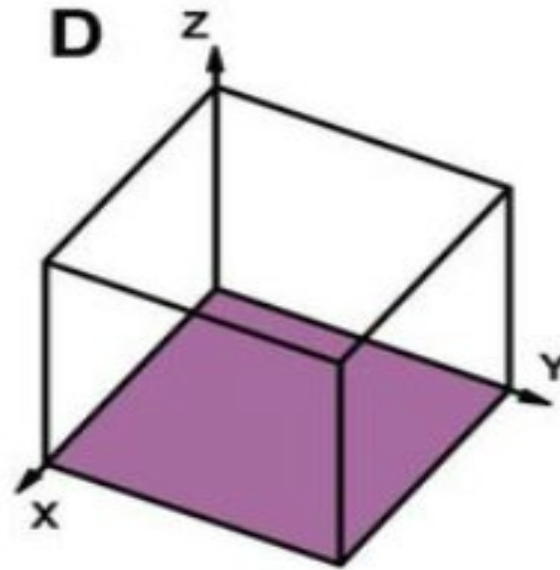
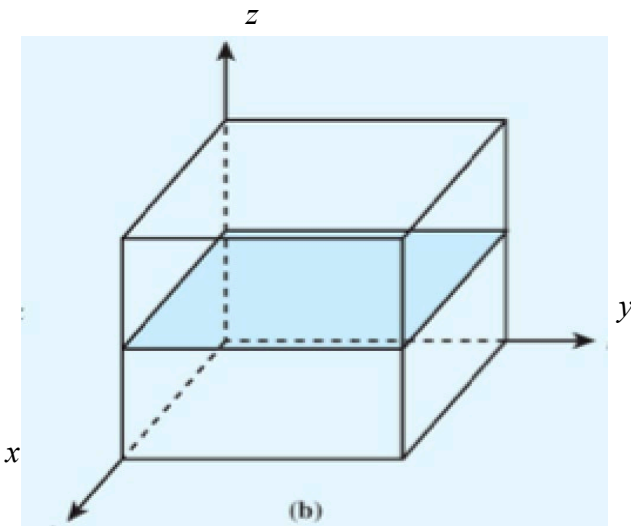
Crystallographic Planes

<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	$1/2$	1	$3/4$
2. Reciprocals	$1/1/2$	$1/1$	$1/3/4$
	2	1	$4/3$
3. Reduction	6	3	4
4. Miller Indices	(634)		



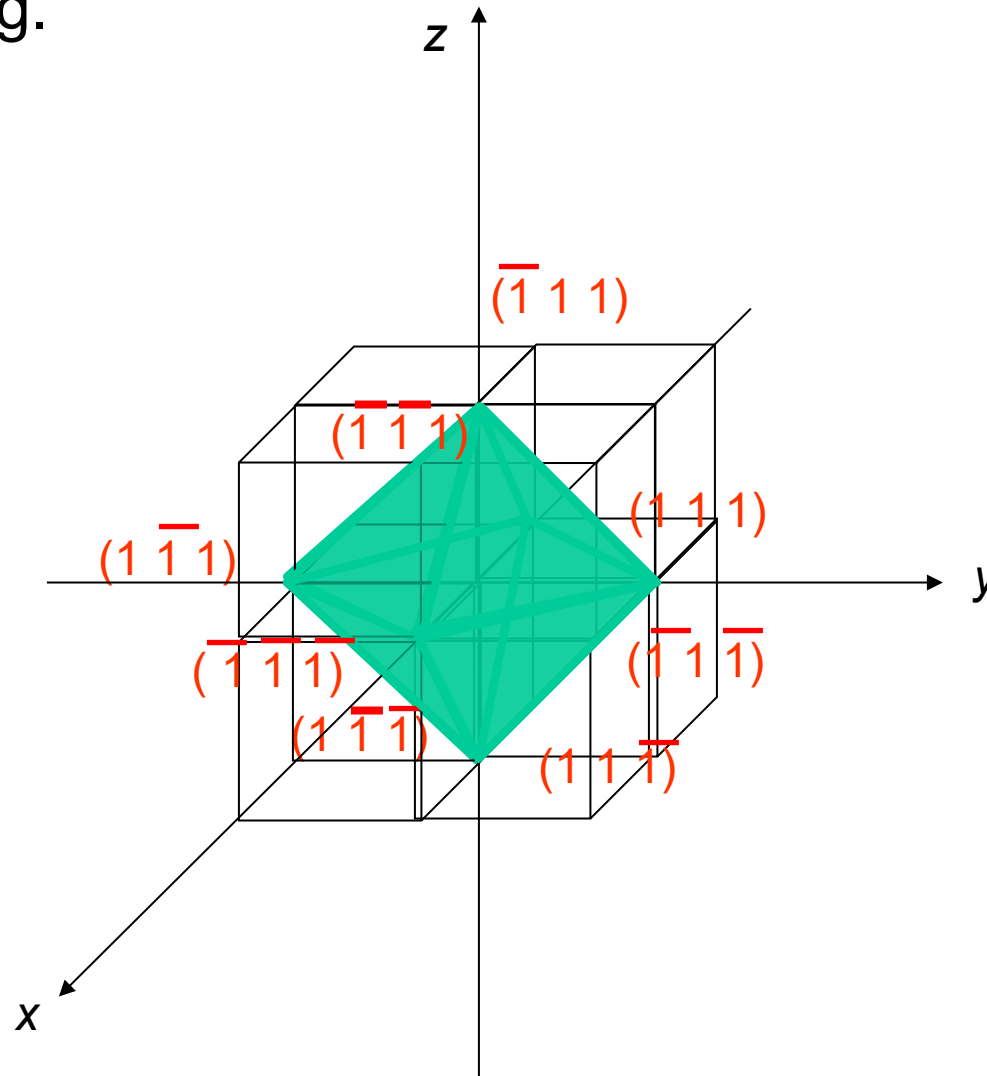
Examples

- What are the Miller indices of the planes shown?



Family of Planes $\{hkl\}$:

Contains all those planes that are crystallographically equivalent, i.e. having the same atomic packing.

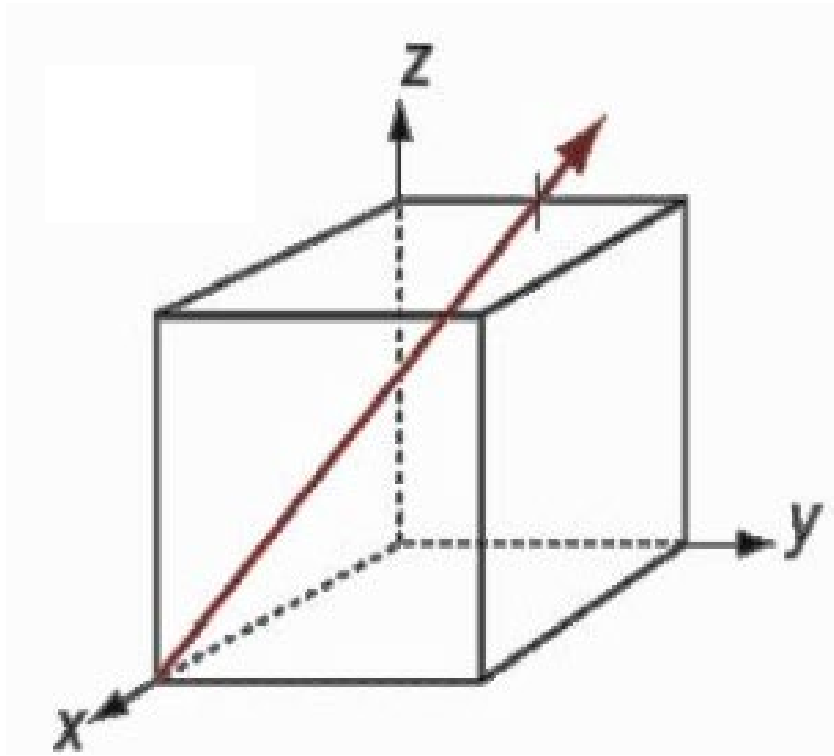


Cubic Unit Cell Plane Family $\{1\ 1\ 1\}$

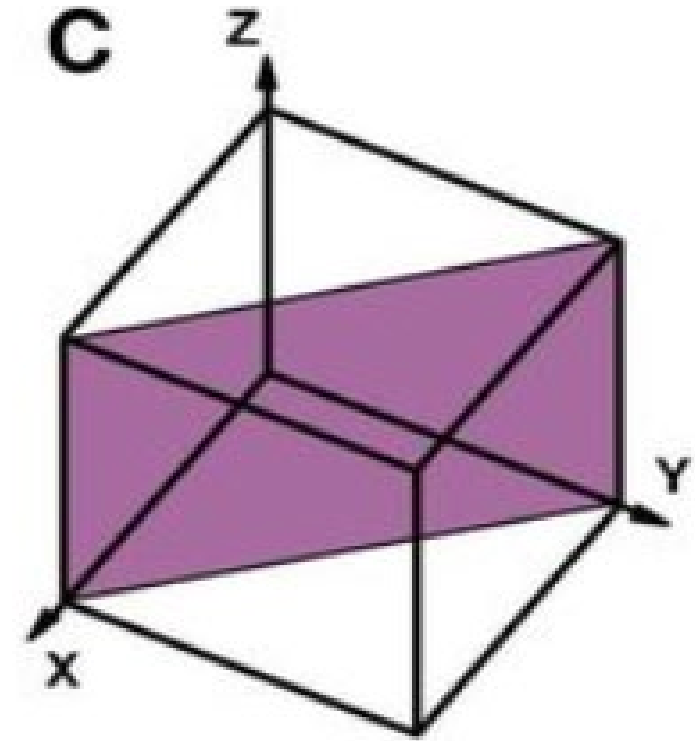
Notice:
Families of planes
have curly brackets

See if you can solve these examples:

Find the directional index:



Find the planar index:



Millers Indices Summary

<u>Type</u>	<u>Format</u>	<u>Ex</u>
• Point Coordinates:	$x\ y\ z$	111
• Directional Indices	$[x\ y\ z]$	[111]
• Families of Directions	$\langle u\ v\ w \rangle$	$\langle 111 \rangle$
• Planal Indices	$(x\ y\ z)$	(111)
• Families of Planes	$\{h\ l\ k\}$	{111}